

Experiment 6: Breadth First Search (BFS)

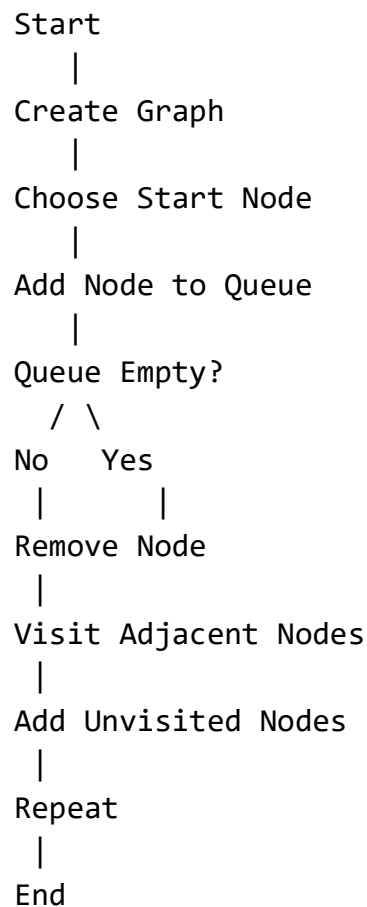
Aim

To implement the Breadth First Search (BFS) algorithm in Python for graph traversal.

Objective

- To traverse all vertices level by level.
- To find the shortest path in an unweighted graph.

Flowchart



Algorithm

1. Create a graph using an adjacency list.
2. Select the starting node.
3. Add the starting node to the queue.
4. Mark it as visited.
5. Remove a node from the queue.
6. Visit all unvisited adjacent nodes.
7. Add them to the queue.
8. Repeat until the queue becomes empty.

Program

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()
queue = deque(['A'])

while queue:
    node = queue.popleft()
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        queue.extend(graph[node])
```

Output

A B C D E F

Result

The BFS traversal was performed successfully.

Conclusion

The Breadth First Search algorithm visits nodes level by level using a queue.